B.E. Comp SEM VIII R-19 'C' 04/12/2024

Sub Code - 52774 Subject : Applied Data Science

(3 Hours) Total Marks 80

N B

1) Question **number 1** is compulsory
2) Attempt **any three** out of the remaining **five questions**.
3) Assume suitable data if **necessary** and justify the assumptions.
4) The figures to the **right** indicate full marks

Q1

A Explain in brief the taxonomy of time series forecasting **5**

B Explain in brief the objectives of Data Exploration **5**

C State the reasons for the outliers occurring in the dataset **5**

D What is anomaly detection? Explain the process of anomaly detection **5**

Q2 A Discuss the working of the ARIMA model in detail **10**

B Explain the DBSCAN algorithm to detect outliers. Give the advantages and disadvantages of the algorithm **10**

Q3 A Explain SMOTE in detail **10**

B Explain the Data Science Process **10**

Q4 A Find Bowley's coefficient of skewness for the following series. **10**

2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22

B State the importance of Data Visualization. State the purpose of scatter plots, quartile plots, bubble charts, density chart **10**

Q5 A Find the coefficient of skewness from the data given below **10**

Size	3	4	5	6	7	8	9	10
Frequency	7	10	14	35	102	136	43	8

B Describe how the time-series approach is useful for forecasting the demand for a product **10**

Q6 A Describe how the predictive modelling can be applied to the House price prediction recommendation **10**

B Explain the significance of Volume, Dimension and Complexity to Data Science Techniques **10**